
NEXGEN BIOLAB

BIOINFORMATICS PROJECTS

Participant Name	Email Address
Fatimah Al-Mohammed	fatimah0almohammed@gmail.com

TASK | Drug-likeness Screening

Objective: To evaluate the drug-likeness of ligand using Lipinski's Rule of Five via SwissADME.

Method Summary:

Drug-likeness screening assesses whether a compound has physicochemical properties consistent with oral bioavailability. Lipinski's Rule of Five (Ro5) states that a drug-like molecule should have: MW $\leq$ 500 Da, H-bond donors $\leq$ 5, H-bond acceptors $\leq$ 10, and LogP $\leq$ 5. Compounds violating more than one rule are unlikely to be orally bioavailable. SwissADME (swissadme.ch) provides a rapid, web-based implementation of these filters alongside additional ADME predictions.

► Task Steps

1	Search for each of the 5 ligands in PubChem (pubchem.ncbi.nlm.nih.gov) and record the CID and canonical SMILES.
2	Go to SwissADME (http://www.swissadme.ch) and paste all 5 SMILES strings (one per line) into the input field.
3	Run the analysis and navigate to the 'Lipinski' and 'Drug-likeness' sections of the results.
4	Record MW, H-bond donors, H-bond acceptors, LogP, rotatable bonds, and overall Ro5 pass/fail for each compound.
5	Download the SDF files from PubChem for each compound.
6	Open and visualize each compound in PyMOL. Take a screenshot of each structure.
7	Take an annotated screenshot of the SwissADME results page showing all 5 compounds.

Compound Information Table

Complete the table below using data from PubChem.

Compound Name	PubChem CID	Canonical SMILES
1 L-Leucine	6106	<chem>CC(C)C[C@@H](C(=O)O)N</chem>
2 L-Alanine	5950	<chem>C[C@@H](C(=O)O)N</chem>
3 L-Valine	6287	<chem>CC(C)[C@@H](C(=O)O)N</chem>
4 L-Isoleucine	6306	<chem>CC[C@H](C)[C@@H](C(=O)O)N</chem>
5 L-Glutamic acid	33032	<chem>C(CC(=O)O)[C@@H](C(=O)O)N</chem>

Lipinski Rule of Five Results

Fill in all values from SwissADME. Use ✓ or ✗ for each rule, then mark overall Pass/Fail.

Compound Name	MW (≤ 500)	HBD (≤ 5)	HBA (≤ 10)	LogP (≤ 5)	Ro5 Pass?
Leucine	131.17	2	3	-1.52	<u>Yes</u> / No
Alanine	89.09	2	3	-2.96	<u>Yes</u> / No
Valine	117.15	2	3	-2.26	<u>Yes</u> / No
Isoleucine	131.17	2	3	-1.72	<u>Yes</u> / No
Glutamic acid	147.13	3	5	-3.69	<u>Yes</u> / No

PyMOL Structures

Figures

Insert one PyMOL screenshot per compound (5 total). Label each image clearly with the compound name.

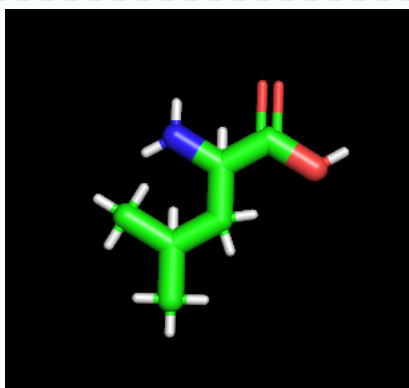


Figure 1. Leucine (PyMOL)

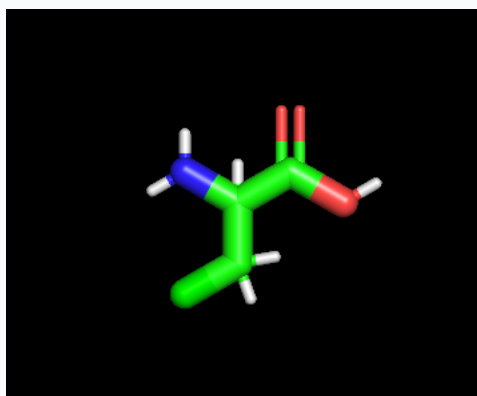


Figure 2. Alanine (PyMOL)

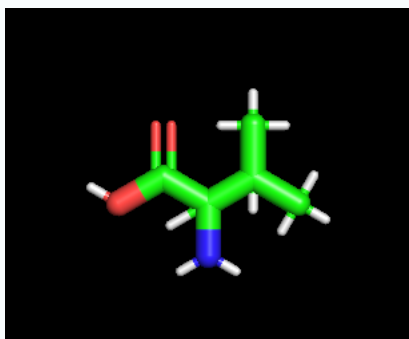


Figure 3. Valine (PyMOL)

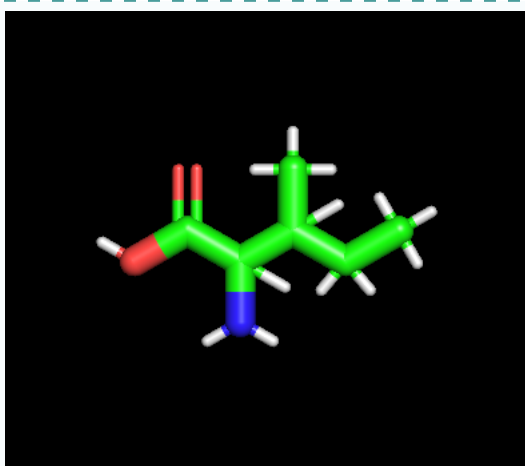


Figure 4. Isoleucine (PyMOL)

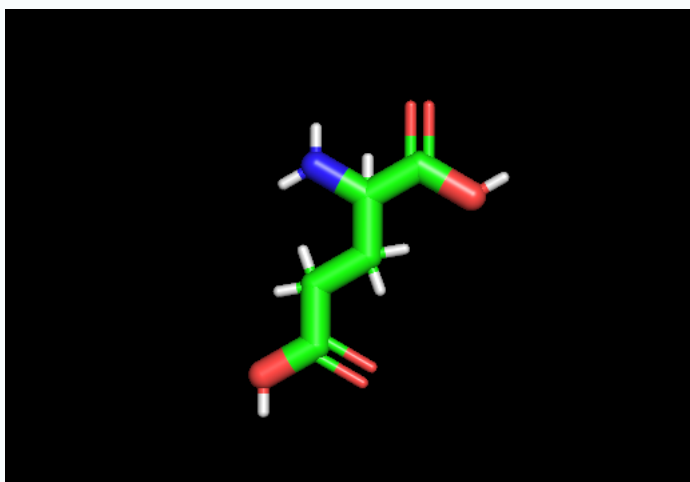


Figure 5. Glutamic Acid (PyMOL)

► Written Justification

Which compounds pass Lipinski's Ro5 and are therefore considered drug-like? Discuss any violations and what they mean for oral bioavailability.

All five compounds (Leucine, Alanine, Valine, Isoleucine, and Glutamic acid) passed Lipinski's Rule of Five with zero violations. Their molecular weights, hydrogen-bond donor and acceptor counts, and LogP values were all within the recommended limits. Therefore, all five compounds meet the Lipinski criteria for drug-likeness. No Lipinski violations were observed, suggesting that none of the compounds has a physicochemical property that would be expected to significantly reduce oral bioavailability based on the Ro5 criteria.

